

Recurring-Regime Reward-Free Exploration: Theory Synopsis

Protocol. An episodic finite MDP has one of finitely many transition kernels. Its hidden kernel is constant within an episode and follows an unknown piecewise-constant schedule across training episodes. Training and q deployment-diagnostic episodes contain no rewards. The deployment kernel is fixed and previously seen. Only after diagnosis is an arbitrary reward $r \in \mathcal{R}_1$, with pathwise total reward at most one, revealed.

Segmentation interface. On an event of probability $1 - \delta_{\text{seg}}$, accepted episodes are pure, recurrences are matched up to one label permutation, at most B episodes at each end of every true block are quarantined, every mode meets its RFE quota, and all confidence sets cover jointly under adaptive collection. This is a contract, not a proved detector theorem.

Theorem 1 (Conditional modular bound). *If segmentation succeeds, every per-mode stationary RFE planner has uniform error η with failure δ_{rfe} , and deployment identification fails with probability at most δ_{id} , then*

$$\Pr \left\{ \sup_{r \in \mathcal{R}_1} V_{P(Z)}^*(r) - V_{P(Z)}^{\hat{\pi}_r}(r) \leq \eta \right\} \geq 1 - \delta_{\text{seg}} - M_\star \delta_{\text{rfe}} - \delta_{\text{id}}.$$

Proof. Intersect segmentation, all $M_\star$ RFE success events, and correct identification. A union bound gives the probability; on the intersection the selected planner is uniformly η -optimal for the deployment kernel. $\square$

Theorem 2 (Binary diagnostic lower bound). *For any adaptive diagnostic strategy σ , pair m, ℓ , and test,*

$$\Pr_m(\ell) + \Pr_\ell(m) \geq \frac{1}{2} \exp\{-D_{\text{KL}}(\mathbb{P}_m^\sigma \| \mathbb{P}_\ell^\sigma)\}.$$

Proof. The minimum sum of testing errors is $1 - \text{TV}$. Apply the Bretagnolle–Huber inequality and $1 - \sqrt{1 - x} \geq x/2$. $\square$

Limits. With no deployment observations, two deterministic modes with reversed good actions force worst-case loss at least $1/2$, even if both models were learned exactly. A single-mode stationary RFE lower bound embeds into any one mode, but multiplying it by the number of modes requires a separate adaptive direct-sum proof. Detector-specific segmentation, robust estimated-model diagnosis, the direct sum, and minimax matching are explicitly unproved targets.

Full details. See `formal_model.md`, `algorithm.md`, `theorem_targets.md`, `proof_roadmap.md`, and `bibliography_ledger.md`.